

Discovering Frequent Closed Embedded Sub-DAGs in Spatio-Temporal Event Data

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Abstract

We propose a novel approach to mine patterns in spatio-temporal event data based on discovering frequent closed embedded sub-Directed Acyclic Graphs (DAGs). In our method, event instances are represented as nodes labelled by event types, while edges capture spatio-temporal following relationships. We formally define the considered class of patterns and justify focusing on closed sub-DAGs as compact and non-redundant representations of recurring interaction patterns. We implement the DigDag algorithm for mining such patterns and experimentally compare its efficiency with two related approaches: propagation pattern mining using the SLEUTH algorithm and Cascading Spatio-Temporal Pattern mining using the CSTPM algorithm. The experimental results demonstrate that our approach is substantially more efficient under comparable parameter settings.

Introduction and Motivation

Discovering spatio-temporal patterns has long attracted attention in areas such as crime analysis, transportation, weather, and epidemiology. In this work, we focus on crime event data, where each event instance is described by its location, occurrence time, and event type.

Our objective is to uncover relationships between different types of crimes. In particular, we ask whether the occurrence of one crime type at a given location is followed by subsequent occurrences of other crime types in the same spatial area and within a given temporal window.

We model such relationships using directed acyclic graphs. Nodes correspond to individual event instances, node labels denote event types, and edges represent the spatio-temporal following relationship. In this setting, a frequent sub-DAG represents a recurring interaction pattern among different event types. By focusing on closed patterns, we eliminate redundancy and retain only the most informative structures.

Main Contributions

- ▶ We propose an approach for discovering frequent closed embedded sub-DAGs in spatio-temporal event data.
- ▶ We define the pattern model, support notion, and closure property.
- ▶ We use the DigDag algorithm to mine these patterns from crime event instances.
- ▶ We compare the approach experimentally with propagation patterns mined by SLEUTH and CSTPs mined by CSTPM.
- ▶ We provide qualitative analysis of selected discovered patterns.

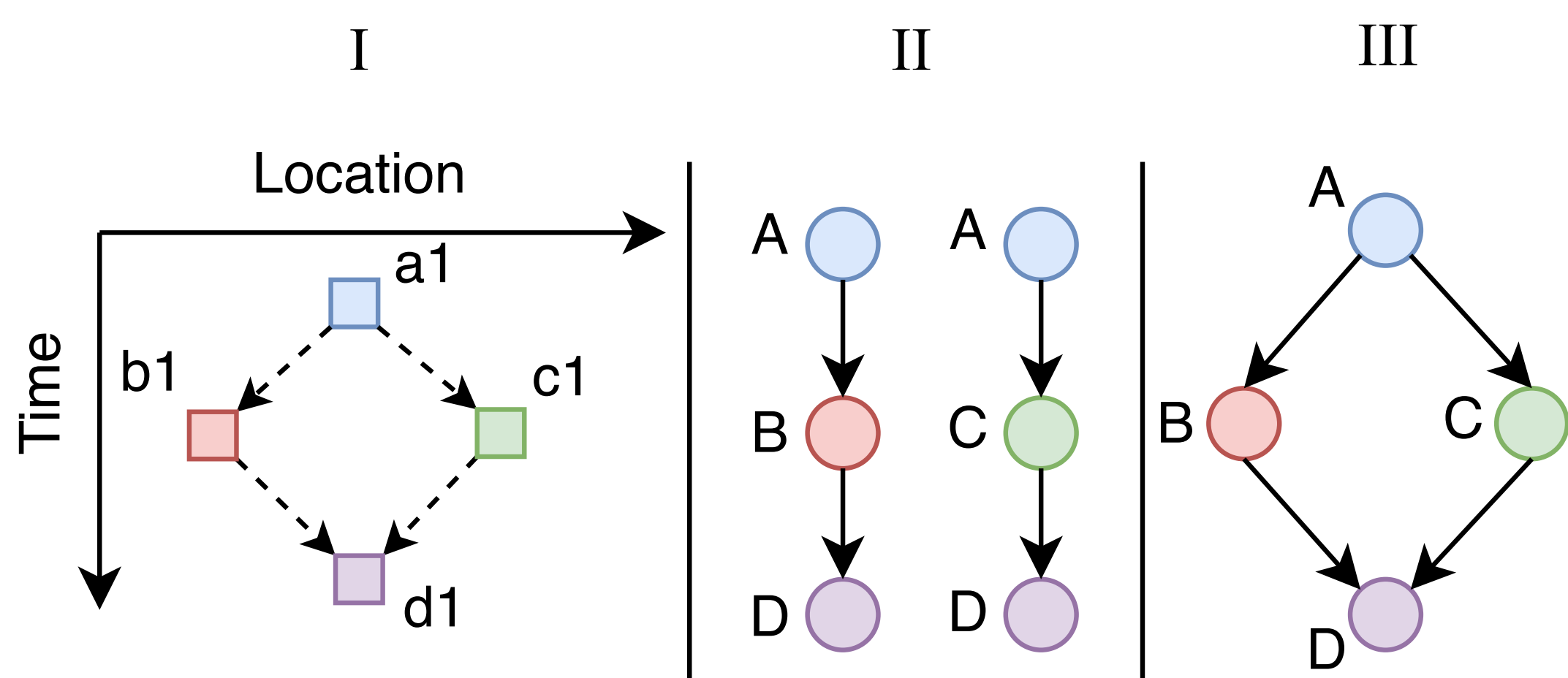
Related Work

Spatio-temporal Sequential Patterns (STSPs) represent relationships as sequences of event types and are evaluated using participation-based measures. Cascading Spatio-Temporal Patterns (CSTPs) extend this idea to partially ordered sets of event types and use the Cascading Participation Index (CPI). Propagation patterns, in turn, are defined as frequent embedded subtrees in forests derived from event data.

Our approach differs in that it focuses on *frequent closed sub-DAGs* rather than sequences or trees. We argue that DAGs are more natural for modelling spatio-temporal event data because they can represent situations in which one event depends on several prior events.

Why DAGs Instead of Trees?

A tree representation may lose information when an event has more than one meaningful predecessor. In contrast, a DAG preserves such multi-parent dependencies. This is important in spatio-temporal event analysis, where an event of one type may be associated with the simultaneous earlier occurrence of several other event types.



Spatio-Temporal Following Relationship

Let $\mathbf{D}$ be the set of event instances and $\mathbf{F}$ the set of event types. Each event instance $i \in \mathbf{D}$ has:

$$i.location, \quad i.time, \quad i.F.$$

For two event instances i_1 and i_2 , we say that i_2 follows i_1 ($i_1 \prec i_2$) if:

$$\text{dist}(i_1.location, i_2.location) \leq R \quad \text{and} \quad (i_2.time - i_1.time) \in (0, T].$$

Here, R is a spatial radius threshold and T is a temporal threshold. In our experiments, distances between geographic coordinates are computed using the Haversine formula.

DAG Model and Pattern Definitions

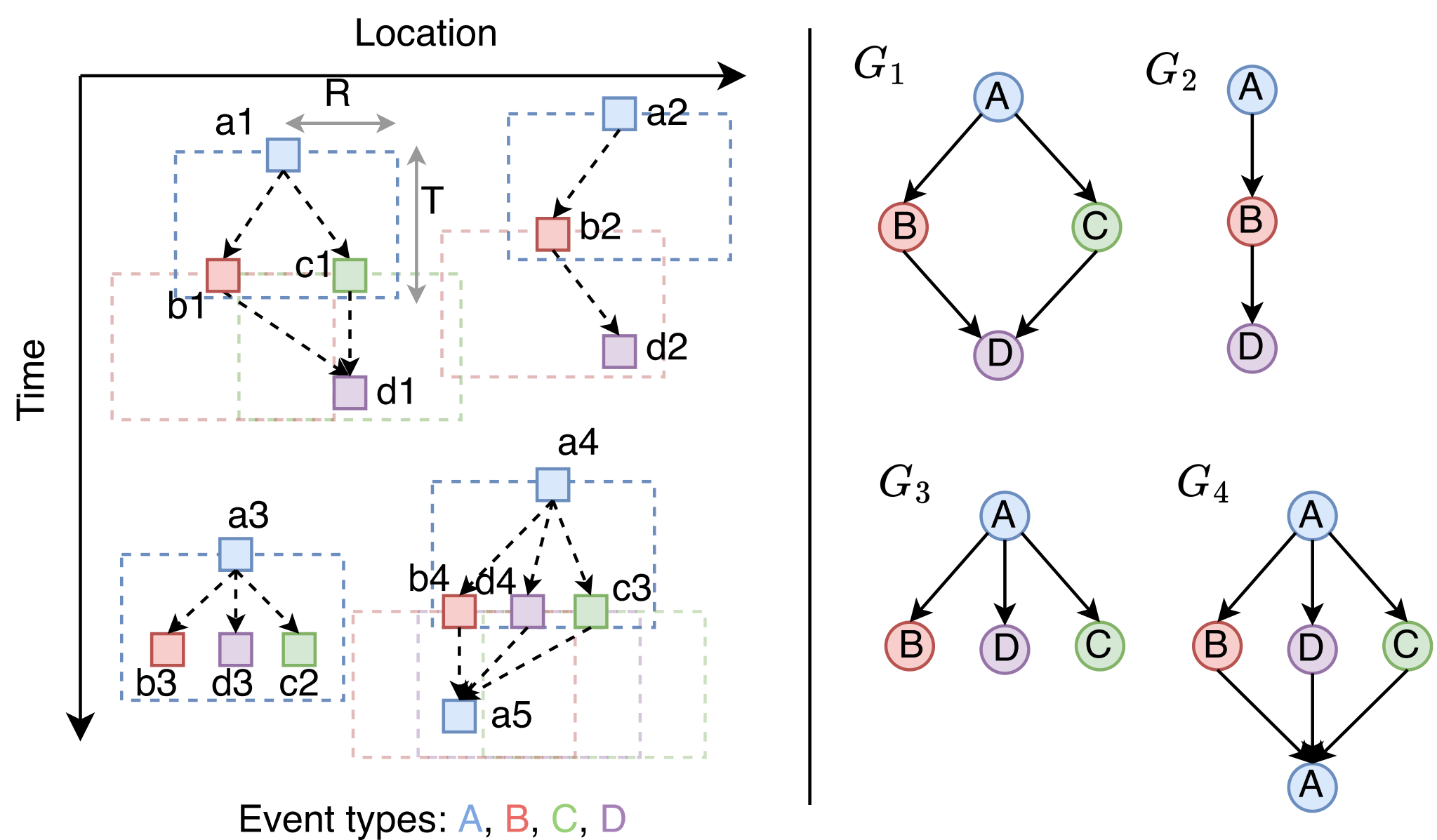
Given the dataset $\mathbf{D}$ and the following relation $\prec$, we define a directed acyclic graph

$$G = (V, E),$$

where

$$V = \{i \mid i \in \mathbf{D}\}, \quad (i_1, i_2) \in E \iff i_1 \prec i_2.$$

Each node is labelled by its event type.



An *embedded sub-DAG* of a DAG $G = (V, E)$ is a graph $G' = (V', E')$ such that $V' \subseteq V$ and each edge $(u, v) \in E'$ corresponds to a directed path from u to v in G .

A sub-DAG is frequent if its support is at least minSup :

$$\text{sup}(G') = \frac{|\{G \in \text{DAGs} \mid G' \text{ is embedded in } G\}|}{|\text{DAGs}|}.$$

A frequent embedded sub-DAG is *closed* if there is no super-pattern with the same support. This yields a compact, non-redundant representation of the frequent pattern space.

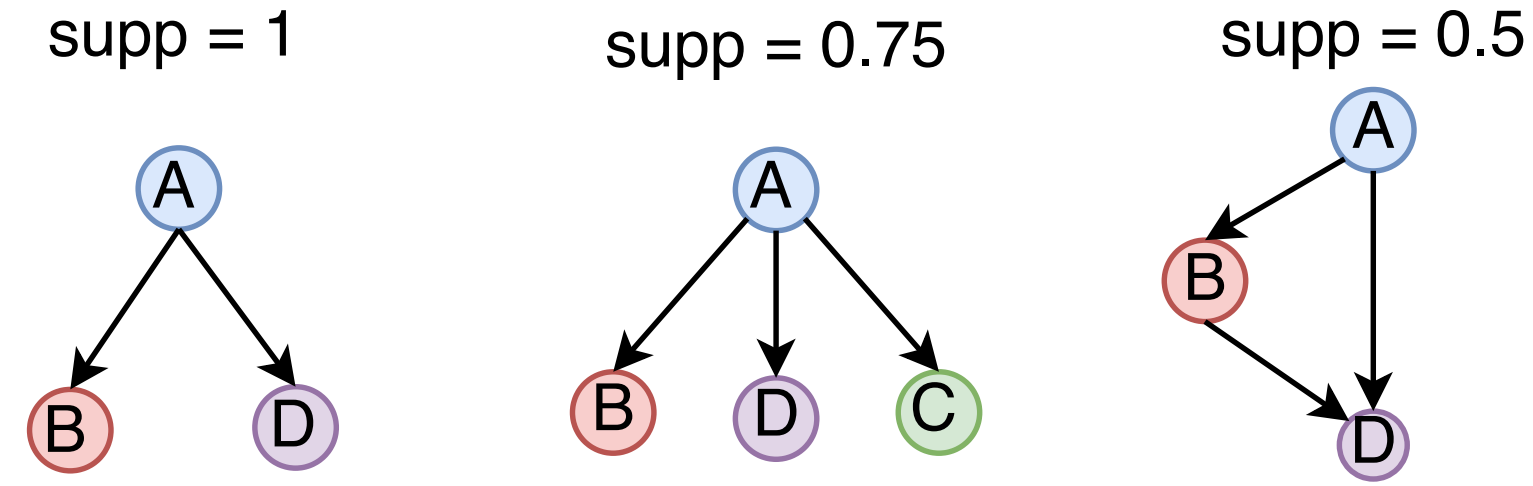
Mining Procedure

The task addressed in this work is to discover all frequent closed embedded sub-DAGs for a given input dataset $\mathbf{D}$ and parameters R , T , and minSup .

For data mining, we employ the DigDag algorithm. The procedure consists of three main steps:

1. transform the event dataset into a set of DAGs,
2. decompose each DAG into tiles,
3. perform closed frequent itemset mining over the generated tiles.

For each DAG, every non-leaf node is treated as a root of a tile consisting of that node and all its descendants. We implemented DigDag in Python using the *networkx* library.



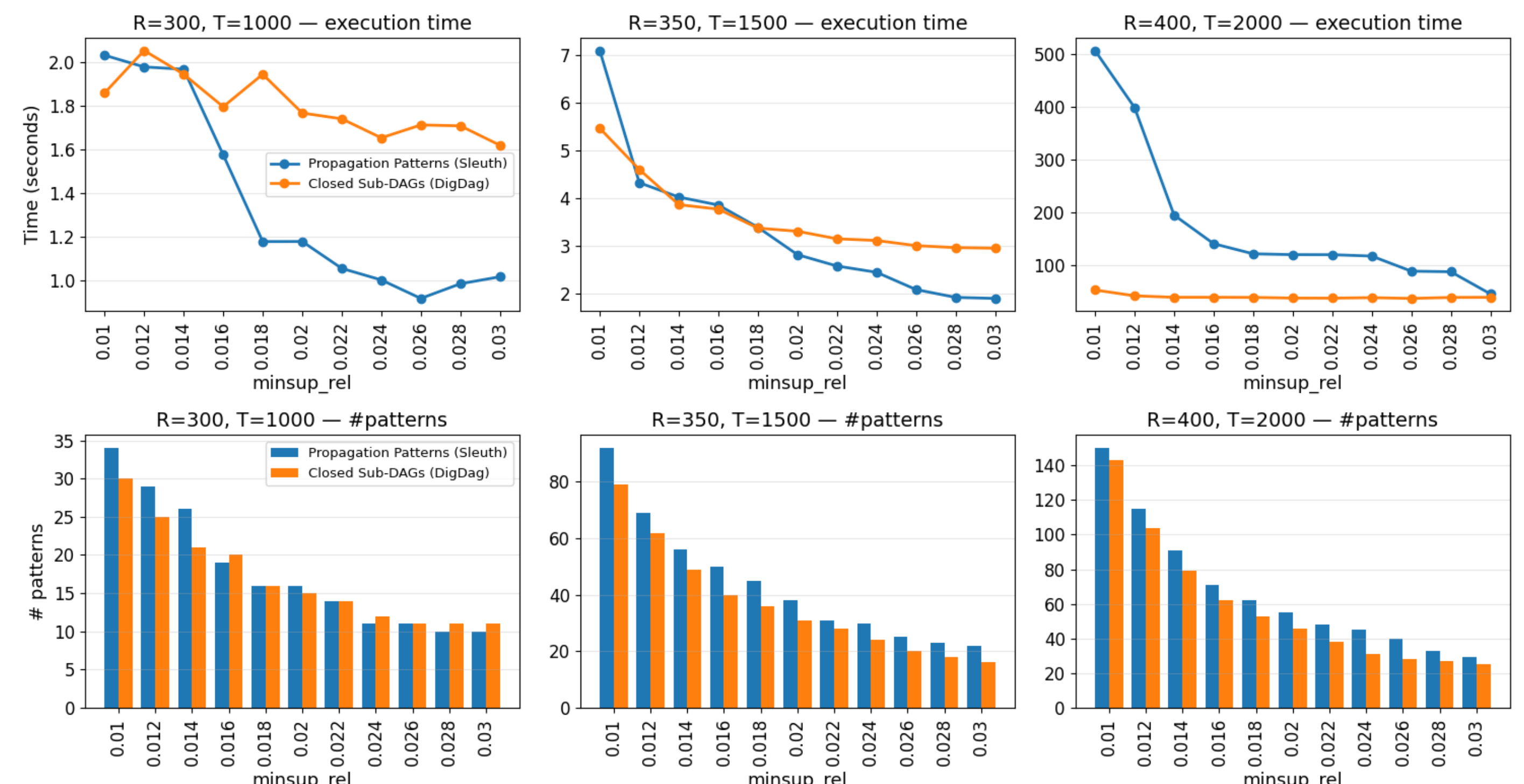
Experimental Setup

We conducted experiments on the *Boston Crime Incidents Reports Dataset*. Only incidents from 01.01.2014 to 31.12.2014 were used. The dataset contains 26 crime event types, such as *auto theft*, *simple assault*, and *vandalism*. We used incident location, time, and type as input attributes.

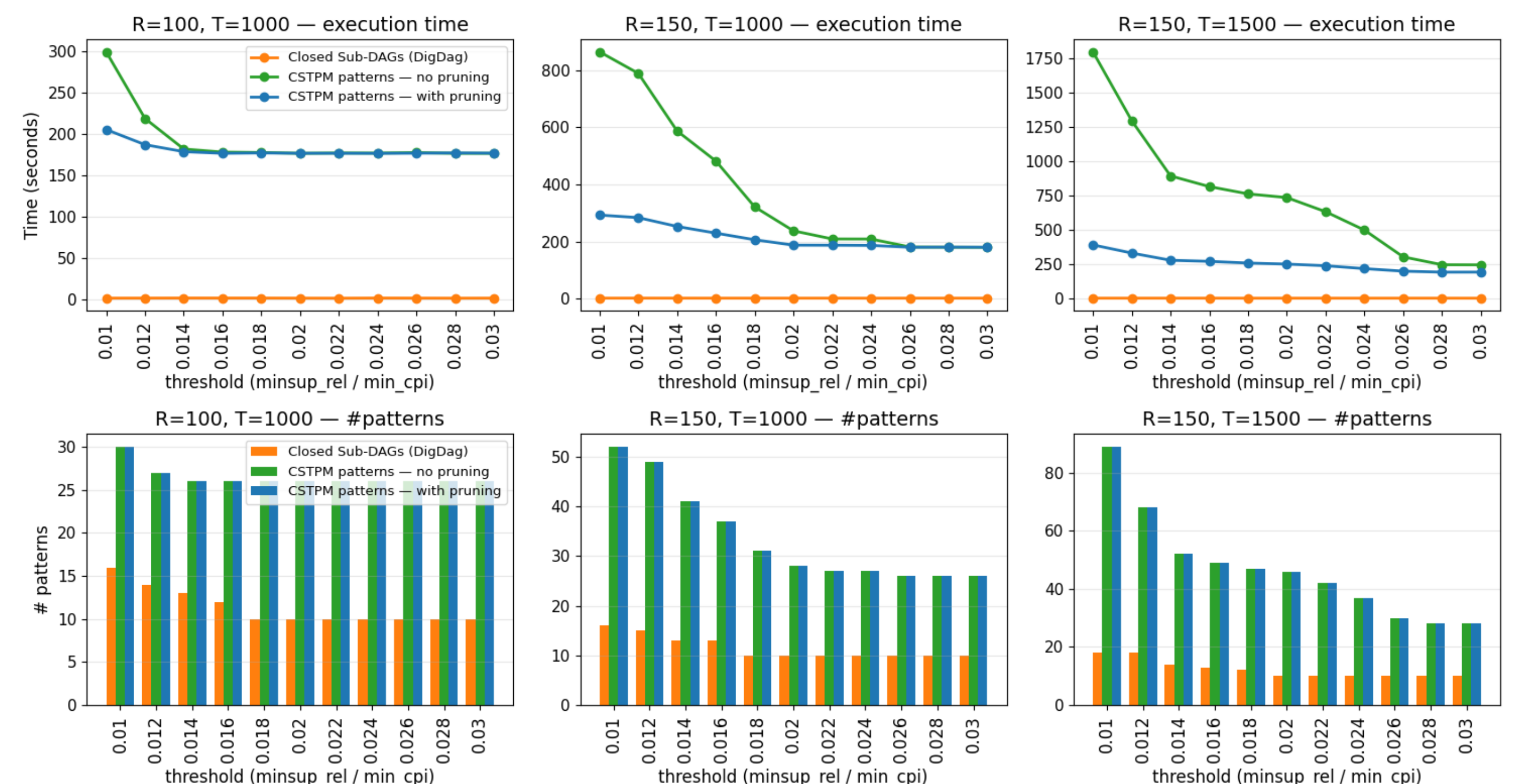
Quantitative Results

We compared our approach with propagation patterns mined using SLEUTH and with CSTPs mined using CSTPM.

For SLEUTH, each DAG had to be transformed into an arborescence because SLEUTH requires tree input. The results show that for smaller parameter settings propagation patterns may be mined slightly faster, but as R and T increase, DigDag becomes significantly more efficient and remains much more stable with respect to varying values of minSup .

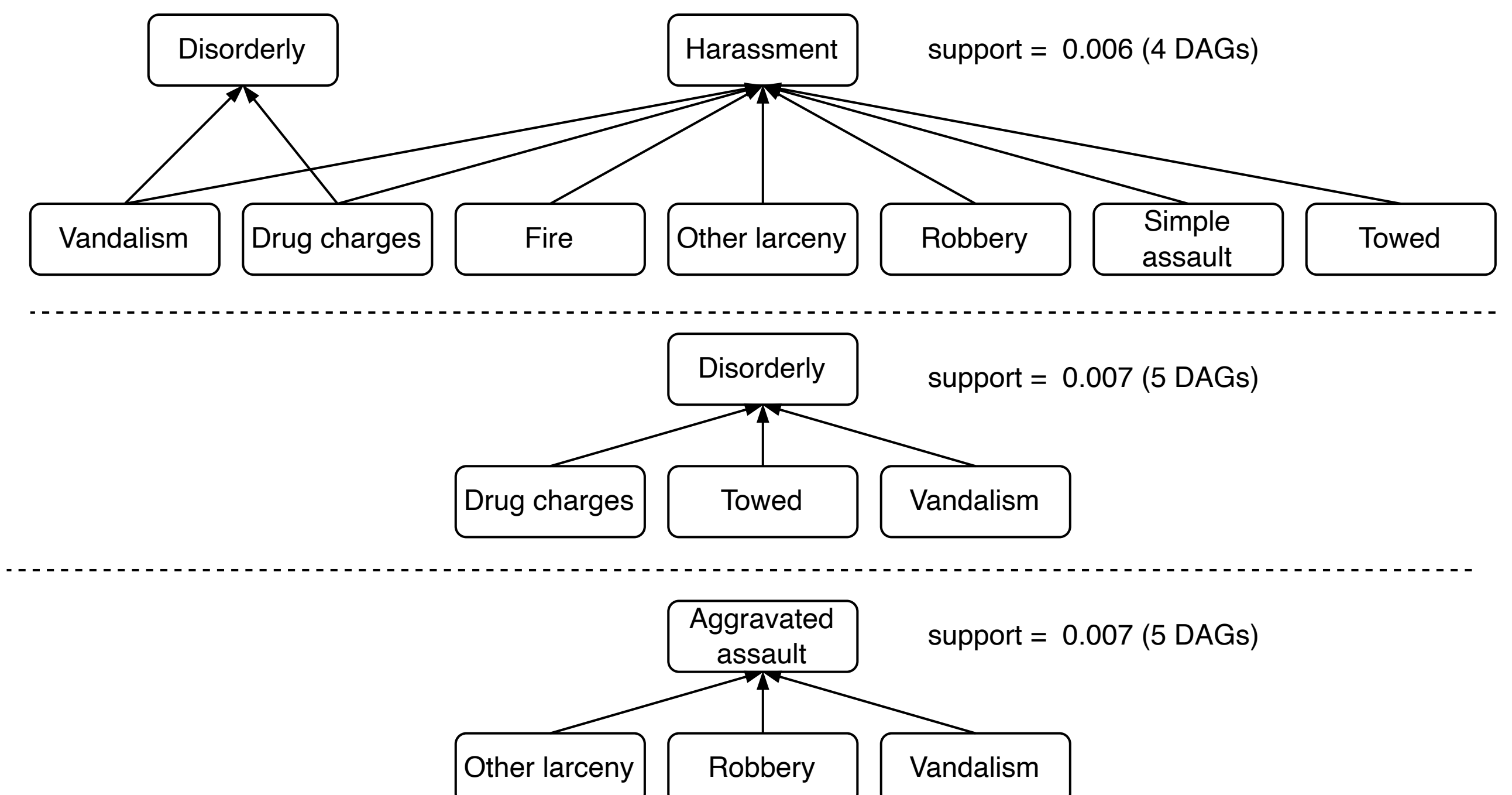


We also compared DigDag with CSTPM. CSTPM performs expensive candidate generation and CPI computation, even when pruning strategies are applied. In our experiments, DigDag was clearly faster.



Qualitative Analysis and Conclusions

For qualitative analysis, we used parameters $R = 400$ meters and $T = 14400$ minutes (10 days), obtaining 626 input DAGs and 1321 frequent patterns for $\text{minSup} = 0.005$.



The discovered patterns suggest potentially meaningful relationships among crime types. For example, events of type *Disorderly* may in some cases be preceded by simultaneous occurrences of *Drug Charges*, *Towed*, and *Vandalism*. Such observations require further validation, but they indicate that frequent closed sub-DAGs may play a useful supporting role in the analysis of criminal activity.

Overall, we conclude that frequent closed embedded sub-DAGs provide a natural and expressive model of spatio-temporal event interactions, while DigDag offers substantially higher efficiency than SLEUTH and CSTPM under comparable settings.